
Homework 5

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Chapter 8

Section 8.5

Problem 8

In a survey of 270 college students, it is found that 64 like brussels sprouts, 94 like broccoli, 58 like cauliflower, 26 like both brussels sprouts and broccoli, 28 like both brussels sprouts and cauliflower, 22 like both broccoli and cauliflower, and 14 like all three vegetables. How many of the 270 students do not like any of these vegetables?

Let $\mathbf{N} = 270$

Let $\mathbf{A} =$ Brussels Sprouts

Let $\mathbf{B} =$ Broccoli

Let $\mathbf{C} =$ Cauliflower

Then,

$$\begin{aligned}\mathbf{A} &= 64, \mathbf{B} = 94, \mathbf{A} = 58 \\ (\mathbf{A}, \mathbf{B}) &= 26, (\mathbf{A}, \mathbf{C}) = 28, (\mathbf{B}, \mathbf{C}) = 22 \\ (\mathbf{A}, \mathbf{B}, \mathbf{C}) &= 14\end{aligned}$$

We want to know what $\overline{N}$ is.

Therefore we need to compute:

$$\overline{N} = N - \sum_{1 \leq i \leq t} N(c_i) + \sum_{1 \leq i < j \leq t} N(c_i c_j) - \sum_{1 \leq i < j < k \leq t} N(c_i c_j c_k)$$

$$\begin{aligned}\overline{N} &= 270 - (64 + 94 + 58) + (26 + 28 + 22) - (14) \\ &= 270 - 216 + 76 - 14 \\ &= 116\end{aligned}$$

Thus, the total amount of students that don't like any of the vegetables is: **116**

Section 8.6

Problem 4

Find the number of solutions of the equation $x_1 + x_2 + x_3 + x_4 = 17$, where $x_i, i = 1, 2, 3, 4$ are non-negative integers such that $x_1 \leq 3, x_2 \leq 4, x_3 \leq 5$, and $x_4 \leq 8$.

Let a solution have a property P_1 if $x_1 > 3$, P_2 if $x_2 > 4$, P_3 if $x_3 > 5$, and P_4 if $x_4 > 8$

The number of total solutions is:

$$\begin{aligned}
N(P'_1 P'_2 P'_3 P'_4) &= N - N(P_1) - N(P_2) - N(P_3) - N(P_4) \\
&\quad + N(P_1 P_2) + N(P_1 P_3) + N(P_1 P_4) + N(P_2 P_3) + N(P_2 P_4) + N(P_3 P_4) \\
&\quad - N(P_1 P_2 P_3) - N(P_1 P_2 P_4) - N(P_1 P_3 P_4) - N(P_2 P_3 P_4) \\
&\quad + N(P_1 P_2 P_3 P_4)
\end{aligned}$$

Then we can compute each value individually before plugging into the formula:

$$\begin{aligned}
N &= \text{Total solutions} C(4 + 17 - 1, 17) = \binom{20}{17} \\
N(P_1) &= \text{Solutions with } x_1 \geq 4 \ C(4 + 13 - 1, 13) = \binom{16}{13} \\
N(P_2) &= \text{Solutions with } x_2 \geq 5 \ C(4 + 12 - 1, 12) = \binom{15}{12} \\
N(P_3) &= \text{Solutions with } x_3 \geq 6 \ C(4 + 11 - 1, 11) = \binom{14}{11} \\
N(P_4) &= \text{Solutions with } x_4 \geq 9 \ C(4 + 8 - 1, 8) = \binom{11}{8} \\
N(P_1 P_2) &= \text{Solutions with } x_1 \geq 4 \text{ and } x_2 \geq 5 = C(4 + 8 - 1, 8) = \binom{11}{8} \\
N(P_1 P_3) &= \text{Solutions with } x_1 \geq 4 \text{ and } x_3 \geq 6 = C(4 + 7 - 1, 7) = \binom{10}{7} \\
N(P_1 P_4) &= \text{Solutions with } x_1 \geq 4 \text{ and } x_4 \geq 9 = C(4 + 4 - 1, 4) = \binom{7}{4} \\
N(P_2 P_3) &= \text{Solutions with } x_2 \geq 5 \text{ and } x_3 \geq 6 = C(4 + 6 - 1, 6) = \binom{9}{6} \\
N(P_2 P_4) &= \text{Solutions with } x_2 \geq 5 \text{ and } x_4 \geq 9 = C(4 + 3 - 1, 3) = \binom{6}{3} \\
N(P_3 P_4) &= \text{Solutions with } x_3 \geq 6 \text{ and } x_4 \geq 9 = C(4 + 2 - 1, 2) = \binom{5}{2} \\
N(P_1 P_2 P_3) &= \text{Solutions with } x_1 \geq 4 \text{ and } x_2 \geq 5 \text{ and } x_3 \geq 6 = C(4 + 2 - 1, 2) = \binom{5}{2} \\
N(P_1 P_2 P_4) &= \text{Solutions with } x_1 \geq 4 \text{ and } x_2 \geq 5 \text{ and } x_4 \geq 9 = 0 \\
N(P_1 P_3 P_4) &= \text{Solutions with } x_1 \geq 4 \text{ and } x_3 \geq 6 \text{ and } x_4 \geq 9 = 0 \\
N(P_2 P_3 P_4) &= \text{Solutions with } x_2 \geq 5 \text{ and } x_3 \geq 6 \text{ and } x_4 \geq 9 = 0 \\
N(P_1 P_2 P_3 P_4) &= \text{Solutions with } x_1 \geq 4 \text{ and } x_2 \geq 5 \text{ and } x_3 \geq 6 \text{ and } x_4 \geq 9 = 0
\end{aligned}$$

After inserting each of these quantities into the formula, we get the following number of solutions:

$$\begin{aligned}
N(P'_1 P'_2 P'_3 P'_4) &= \binom{20}{17} \text{Total Number of Solutions} \\
&\quad - \binom{16}{13} - \binom{15}{12} - \binom{14}{11} - \binom{11}{8} \quad (P_1), (P_2), (P_3), (P_4) \\
&\quad + \binom{11}{8} + \binom{10}{7} + \binom{7}{4} + \binom{9}{6} + \binom{6}{3} + \binom{5}{2} \quad (P_1 P_2), (P_1 P_3), (P_1 P_4), (P_2 P_3), (P_2 P_4), (P_3 P_4) \\
&\quad - \binom{5}{2} - 0 - 0 - 0 \quad (P_1 P_2 P_3), (P_1 P_2 P_4), (P_1 P_3 P_4), (P_2 P_3 P_4) \\
&\quad + 0 \quad (P_1 P_2 P_3 P_4)
\end{aligned}$$

Problem 6

An integer is called **squarefree** if it is not divisible by the square of a positive integer greater than 1. Find the number of **squarefree** positive integers less than 100.

We can re-write this as saying that we want to find numbers that are divisible by the square of any prime number less than 10. Thus, we're interested in numbers that are divisible by $2^2, 3^2, 5^2, 7^2$. As such, we take each number to be a property of our possible solutions.

Using the principle of Inclusion / Exclusion we plug values into the same formula as **Problem 4**, namely:

$$\begin{aligned} N(P'_1 P'_2 P'_3 P'_4) &= N - N(P_1) - N(P_2) - N(P_3) - N(P_4) \\ &\quad + N(P_1 P_2) + N(P_1 P_3) + N(P_1 P_4) + N(P_2 P_3) + N(P_2 P_4) + N(P_3 P_4) \\ &\quad - N(P_1 P_2 P_3) - N(P_1 P_2 P_4) - N(P_1 P_3 P_4) - N(P_2 P_3 P_4) \\ &\quad + N(P_1 P_2 P_3 P_4) \end{aligned}$$

Note that there are 99 integers less than 100, as such it follows that:

$$\begin{aligned} N(P'_1 P'_2 P'_3 P'_4) &= 99 \quad \text{Total number of integers smaller than 100} \\ &\quad - \left\lfloor \frac{99}{2^2} \right\rfloor - \left\lfloor \frac{99}{3^2} \right\rfloor - \left\lfloor \frac{99}{5^2} \right\rfloor - \left\lfloor \frac{99}{7^2} \right\rfloor \\ &\quad (P_1), (P_2), (P_3), (P_4) \\ &\quad + \left\lfloor \frac{99}{2^2 \cdot 3^2} \right\rfloor + \left\lfloor \frac{99}{2^2 \cdot 5^2} \right\rfloor + \left\lfloor \frac{99}{2^2 \cdot 7^2} \right\rfloor + \left\lfloor \frac{99}{3^2 \cdot 5^2} \right\rfloor + \left\lfloor \frac{99}{3^2 \cdot 7^2} \right\rfloor + \left\lfloor \frac{99}{5^2 \cdot 7^2} \right\rfloor \\ &\quad (P_1 P_2), (P_1 P_3), (P_1 P_4), (P_2 P_3), (P_2 P_4), (P_3 P_4) \\ &\quad - \left\lfloor \frac{99}{2^2 \cdot 3^2 \cdot 5^2} \right\rfloor - \left\lfloor \frac{99}{2^2 \cdot 3^2 \cdot 7^2} \right\rfloor - \left\lfloor \frac{99}{2^2 \cdot 5^2 \cdot 7^2} \right\rfloor - \left\lfloor \frac{99}{3^2 \cdot 5^2 \cdot 7^2} \right\rfloor \\ &\quad (P_1 P_2 P_3), (P_1 P_2 P_4), (P_1 P_3 P_4), (P_2 P_3 P_4) \\ &\quad + \left\lfloor \frac{99}{2^2 \cdot 3^2 \cdot 5^2 \cdot 7^2} \right\rfloor \\ &\quad (P_1 P_2 P_3 P_4) \end{aligned}$$

Plugging in the values we have: $N(P'_1 P'_2 P'_3 P'_4) = 99 - (24 + 11 + 3 + 2) + (2 + 0 + 0 + 0 + 0) - (0 + 0 + 0 + 0) + 0 = 61$

To conclude, there are 61 Numbers that are **squarefree** and less than 100.

Chapter 9

Section 9.1

Problem 6

Determine whether the relation R on the set of all real numbers is *reflexive*, *symmetric*, *anti-symmetric*, and/or *transitive*, where $(x, y) \in R$ if and only if

a) $x + y = 0$

¬ **Reflexive** - Because the (x, x) does not exist in the relationship for every value of x .

Symmetric - because if $x + y = 0$, then $y + x = 0$ is also true.

¬ **Anti-Symmetric** - Because (x, y) and (y, x) can be in the relationship, but $x \neq y$.

¬ **Transitive** - suppose we have $x = 1$, $y = -1$, and $z = 1$. Then $x + y = 0$, $y + z = 0$ but $x + z \neq 0$

b) $x = \pm y$

Reflexive - because (x, x) is in the set for every element of the set.

Symmetric - because x, y and y, x are in the set. ¬ **Anti-Symmetric** - $(x, y) \wedge (y, x)$ is true, but $x \neq y$ when the signs are different.

Transitive - because $x \pm y \pm z$.

c) $x - y$ is a rational number.

Reflexive - because $x - x$ can always be represented as a rational number.

Symmetric - because for every $x - y$ that is rational, $y - x$ will also be rational.

$\neg$ **Anti-Symmetric** - because for every relationship of x, y and y, x , the implication $x = y$ is not true.

$\neg$ **Transitive** - because if $x = 1$, $y = \frac{1}{2}$, and $z = 2$, then $x - y$ is rational, $y - z$ is rational, but $x - z$ is not rational.

d) $x = 2y$

$\neg$ **Reflexive** - $(2, 1)$ will be in the set, but $(2, 2)$ will not be in the set.

$\neg$ **Symmetric** - $(2, 1)$ will be in the set, but $(1, 2)$ will not be in the set.

Anti-Symmetric - This is vacuously true since $(x, y) \wedge (y, x)$ is not true.

$\neg$ **Transitive** - $(2, 1)$ is in the set, $(1, \frac{1}{2})$ is in the set, but $2 \neq 2 \cdot \frac{1}{2}$.

e) $xy \geq 0$

Reflexive - because for any $xx \geq 0$, the result is ≥ 0 .

Symmetric - because for any $xy \geq 0$, $yx \geq 0$ is also true.

Anti-Symmetric - because $xy \geq 0$ can be true, but $x \neq y$ can also be true for the same relationship.

Transitive - because the relationship x, y and y, z are in the relationship, and $x \geq z$ for every value.

f) $xy = 0$

Reflexive - because $xx = 0$ is not true for every value of x .

Symmetric - because $xy = 0$, then $yx = 0$ is also true.

$\neg$ **Anti-Symmetric** - because $xy = 0$ does not imply that $x = y$, consider the case when $x = 0$, and $y = 1$. $xy = 0$, but $x \neq y$.

Transitive - because for $x = 0$, $y = 1$, $z = 2$, $xy \geq 0$, $yz \geq 0$, and $xz \geq 0$ are all true for the set.

g) $x = 1$

$\neg$ **Reflexive** - because $x = 1$ for every pair in the relationship, therefore since y can be any other number $(1, 1)$ only occurs once in the set.

$\neg$ **Symmetric** - because we cannot have (x, y) and (y, x) be true unless $y = 1$, and this is only one ordered pair in the set.

Anti-Symmetric - because of the same reason that it is not Symmetric. This implies that $x = y$ is vacuously true.

Transitive - If (x, y) and (y, z) are in the set, then $x = y = 1$.

h) $x = 1$ or $y = 1$.

$\neg$ **Reflexive** - because $x = 1$ for $\mathbb{R}$, which means that it doesn't express ever value, only $(1, 1)$, and the same holds true for $y = 1$.

Symmetric - because for every $(1, y)$ there is a $(x, 1)$ pair in the set.

Anti-Symmetric - because of the statement above, as well as the fact that $x = y$.

Transitive - because $(1, 1)$ and $(1, z)$ are in the set.

Section 9.5

Problem 2

Which of these relations on the set of all people are equivalence relations? Determine the properties of an equivalence relation that the others lack.

a) $(a, b) | a$ and b are the same age

Has an equivalence relation

Reflexive - a is the same age as a .

Symmetric - if a is the same age as b , then b must be the same age as a .

Transitive - if a is the same age as b and b is the same age as c , then a has to be the same age as c .

b) $(a, b) | a$ and b have the same parents

Has an equivalence relation

Reflexive - a has the same parent as a .

Symmetric - if a has the same set of parents as b , then b must have the same set of parents as a .

Transitive - if a has the same set of parents as b , and b has the same set of parents as c , then a and c have the

same parents.

c) $(a, b) | a$ and b share a common parent

Does NOT have an equivalence relation

$\neg$ **Transitive** a can share a common parent with b , but b might share a different common parent with c .

d) $(a, b) | a$ and b have met

Does NOT have an equivalence relation

$\neg$ **Transitive** - a and b have met, b and c have met, but a and c could have not met.

e) $(a, b) | a$ and b speak a common language

Does NOT Have an equivalence relation

$\neg$ **Transitive** - if a and b speak a common language, and b and c speak a different common language, that does not imply that a and c speak the same common language since each language relationship could be different.

Problem 46

Which of these are partitions of the set of real numbers?

a) the negative real numbers, $\{0\}$, the positive real numbers

All three are partitions of the set of real numbers.

b) the set of irrational numbers, the set of rational numbers

The set of rational numbers.

c) the set of intervals $[k, k + 1], k = \dots, -2, -1, 0, 1, 2, \dots$

This is not a partition because for the intervals $[1, 2]$ and $[2, 3]$ the number 2 exists in both sets.

d) the set of intervals $(k, k + 1), k = \dots, -2, -1, 0, 1, 2, \dots$

This is a partition.

e) the set of intervals $(k, k + 1], k = \dots, -2, -1, 0, 1, 2, \dots$

This is a partition.

f) the sets $x + n | n \in \mathbb{Z} \quad \forall x \in [0, 1)$

This is a partition because n cannot be 0, and x can never be a whole number.

Therefore, each set will never contain the same number since we can't have $x = 0, n = 1$, and $x = 1, n = 0$

Section 9.6

Problem 2

Which of these relations on $\{0, 1, 2, 3\}$ are partial orderings? Determine the properties of a partial ordering that the others lack.

a) $\{(0, 0), (1, 1), (2, 2), (3, 3)\}$

This is a partial order.

b) $\{(0, 0), (1, 1), (2, 0), (2, 2), (2, 3), (3, 3)\}$

This is a partial order.

c) $\{(0, 0), (1, 1), (1, 2), (2, 2), (3, 3)\}$

This is not a partial order, because it is not transitive.

d) $\{(0, 0), (1, 1), (1, 2), (1, 3), (2, 2), (2, 3), (3, 3)\}$

This is a partial order.

e) $\{(0, 0), (0, 1), (0, 2), (1, 0), (1, 1), (1, 2), (2, 0), (2, 2), (3, 3)\}$

This is not a partial order because it is not anti-symmetric. $(0, 1)$ and $(1, 0)$ are in the set, but $1 \neq 0$.

Problem 6

Which of these are posets?

a) $(\mathbb{R}, =)$

This is a poset.

b) $(\mathbb{R}, <)$

This is not a poset, because $<$ is not reflexive.

c) $(\mathbb{R}, \leq)$

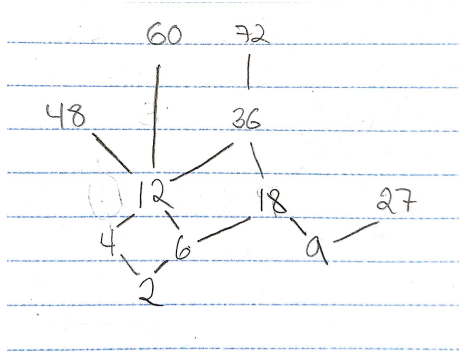
This is a poset.

d) $(\mathbb{R}, \neq)$

This is not a poset, because $\neq$ is not reflexive.

Problem 34

Answer these questions for the poset $(\{2, 4, 6, 9, 12, 18, 27, 36, 48, 60, 72\}, |)$.



a) Find the maximal elements.

27, 48, 60, 72

b) Find the minimal elements.

2, 9

c) Is there a greatest element?

No, there are 3 maximals, which means there cannot be a greatest element.

d) Is there a least element?

No, there cannot more than 1 least element, and this graph has 2, 9

e) Find all upper bounds of $\{2, 9\}$.

18, 36, 72

f) Find the least upper bound of $\{2, 9\}$, if it exists.

18

g) Find all lower bounds of $\{60, 72\}$.

2, 4, 12

h) Find the greatest lower bound of 60, 72, if it exists.

12

Chapter 10

Section 10.2

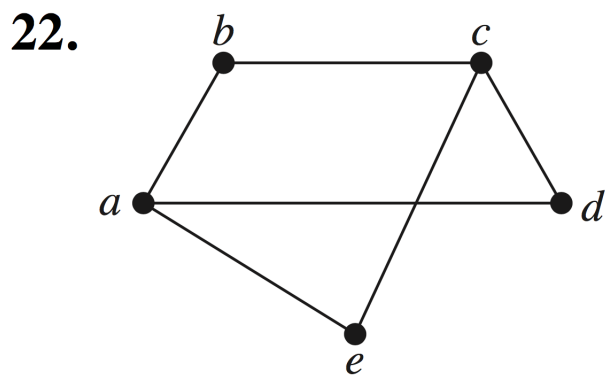
Problem 18

Show that in a simple graph with at least two vertices there must be two vertices that have the same degree.

If the minimum number of degrees for n vertices, is 1, and the maximum is $n - 1$. Then we can guarantee that the graph will have at least 2 vertices with the same number of degrees.

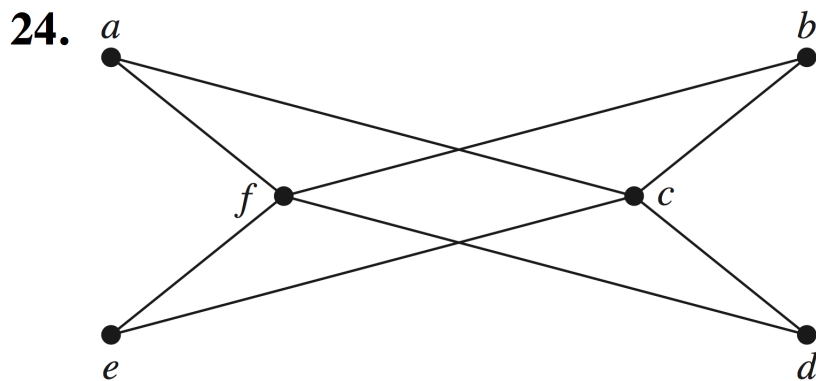
Problem 22

Determine whether the graph is bipartite. You may find it useful to apply Theorem 4 and answer the question by determining whether it is possible to assign either red or blue to each vertex so that no two adjacent vertices are assigned the same color.



This graph is Bipartite.

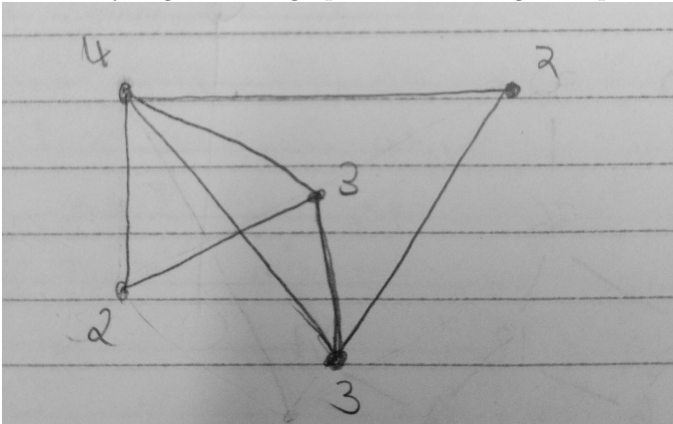
Problem 24



This graph is Bipartite.

Problem 40

How many edges does a graph have if its degree sequence is 4, 3, 3, 2, 2? Draw such a graph.



7 EDGES

Problem 60

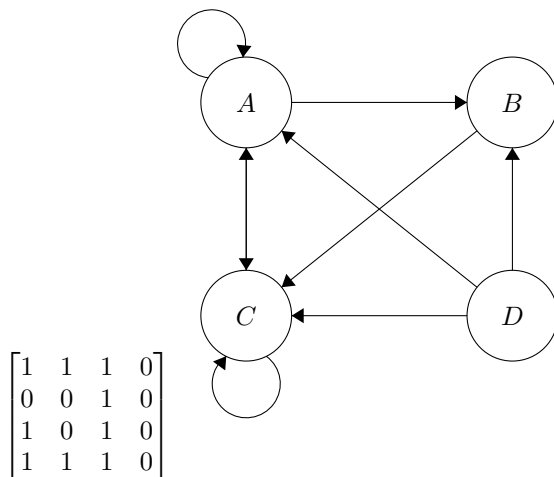
If G is a simple graph with 15 edges and $\overline{G}$ has 13 edges, how many vertices does G have?

$$\begin{aligned}
 |G| + |\overline{G}| &= \frac{n(n-1)}{2} \\
 15 + 13 &= \frac{n(n-1)}{2} \\
 28 &= \frac{n(n-1)}{2} \\
 56 &= n(n-1) \\
 -n^2 + n + 56 &= 0 \\
 (-n-7)(n-8) &= 8
 \end{aligned}$$

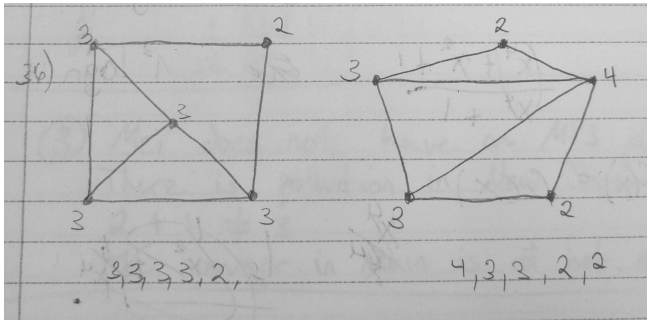
Since -7 does not make sense, the graph G has 8 vertices.

Section 10.3

Problem 12



Problem 36



NOT ISOMORPHIC

Problem 38

ISOMORPHIC

If we count each in-degree for each connecting vertex we can compare both graphs. If the degree sequences for each graph do not match, then the graphs are not Isomorphic.

$$\begin{array}{ll}
 v_1 = 4, 3, 3 & u_1 = 4, 4, 3 \\
 v_2 = 4, 3, 3, 2 & u_2 = 4, 4 \\
 v_3 = 4, 4 & u_3 = 4, 3, 3, 2 \\
 v_4 = 4, 3, 3, 2 & u_4 = 4, 4, 3 \\
 v_5 = 4, 4, 3 & u_5 = 4, 3, 3, 2
 \end{array}$$

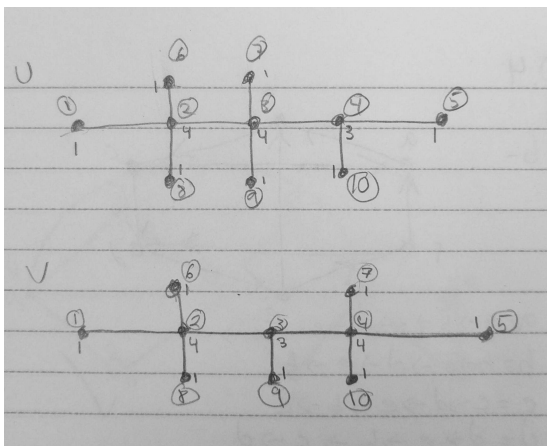
The degree Sequences are equal, so the graphs are Isomorphic.

Problem 40

$$\begin{array}{ll}
 v_1 = 4, 3, 3 & u_1 = 4, 4, 3 \\
 v_2 = 4, 3, 3, 2 & u_2 = 4, 4 \\
 v_3 = 4, 4 & u_3 = 4, 3, 3, 2 \\
 v_4 = 4, 3, 3, 2 & u_4 = 4, 4, 3 \\
 v_5 = 4, 4, 3 & u_5 = 4, 3, 3, 2
 \end{array}$$

NOT ISOMORPHIC

Problem 42



$v_1 = 4$	$u_1 = 4$
$v_2 = 3, 1, 1$	$u_2 = 4, 1, 1$
$v_3 = 4, 4, 1$	$u_3 = 4, 3, 1, 1$
$v_4 = 3, 1, 1, 1$	$u_4 = 4, 2, 1$
$v_5 = 4$	$u_5 = 4$
$v_6 = 4$	$u_6 = 4$
$v_7 = 4$	$u_7 = 4$
$v_8 = 4$	$u_8 = 4$
$v_9 = 3$	$u_9 = 3$
$v_{10} = 4$	$u_{10} = 4$

NOT ISOMORPHIC

Section 10.4

Problem 12

b) Is strongly connected.

Problem 32

The cut vertices are $\{c, d\}$

Problem 34

Problem 31 Does not have cut edges.

Problem 32 The cut edges is (c, b)

Problem 33 The cut edges are $(a, b), (b, c), (c, d), (c, e), (e, i), (i, h)$

Section 10.5

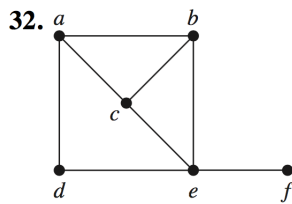
Problem 6

A Euler circuit does not exist, but a Euler path does.

Euler Path: $b \rightarrow a \rightarrow d \rightarrow e \rightarrow f \rightarrow d \rightarrow c \rightarrow i \rightarrow d \rightarrow g \rightarrow i \rightarrow a \rightarrow h \rightarrow i \rightarrow b \rightarrow c$

Problem 32

Not a Hamiltonian Circuit



This graph does not have a hamiltonian cycle because for any path taken, Node e will be traversed twice.

Problem 38

Does the Graph in Exercise 31 have a Hamilton Path. If so, find such a path. If not, give an argument to show why no such path exists.

